Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: Test automation, Hardware-in-the-loop testing, Embedded C, Automotive Ethernet, Linux, MQTT, CAN bus, TCP/IP, Wireshark, MATLAB, Data acquisition, Grafana, Docker, InfluxDB
Data Engineering: Apache Kafka
Languages: SQL, Python, C, C++, C#
Machine Learning and AI:

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements

Projects

CAN Bus Telemetry Logger and Dashboard *Python, Embedded C, MQTT, InfluxDB, Grafana*

- Built a simulated CAN bus data source and a Python gateway that decodes vehicle-style frames and publishes messages over MQTT for near real-time monitoring
- Stored decoded telemetry channels in InfluxDB and assembled Grafana dashboards for session-by-session analysis of sensor channels and system health
- Added configurable alerting on dropped packets and out-of-range channels to practice trackside-style fault diagnosis under time pressure

FPGA-Based Autonomous Vehicle *Xilinx Arty FPGA, MicroBlaze, Vivado, Vitis, AXI, C*

- Built an autonomous line-following vehicle on a Xilinx Arty FPGA with a custom MicroBlaze-based SoC, integrating IR line sensors, ultrasonic sensors, and motor drivers through AXI IP cores in Vivado
- Wrote embedded firmware in C using Vitis for line tracking, obstacle detection, sensor processing, and motor control
- Implemented Quad-SPI flash boot with an SREC bootloader enabling autonomous startup after power-up